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Identification of a Single Missense Mutation in the Adenine Phosphoribosyltransferase (APRT) Gene from Five Icelandic Patients and a British Patient

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Summary

We have completely sequenced the adenine phosphoribosyltransferase (APRT) gene from each of six patients—five (I–V) from Iceland and one (VI) from Britain. Cases I and II shared a common ancestor six and seven generations ago, and cases I and V shared a common ancestor seven generations ago, but cases III and IV were unrelated to the above or to each other, over seven generations. Genomic DNA was amplified by PCR, subcloned into M13mp18, and sequenced. Genomic and PCR-amplified DNAs were also analyzed by restriction-enzyme digestion and Southern blotting. The same missense mutation was identified in all six patients. This mutation leads to the replacement of asp (GAC) by val (GTC), at amino acid position 65. The gene sequences from all patients were otherwise identical to our wild-type sequence. The homozygous nature of the mutation was confirmed by sequencing the PCR product directly. All six patients were homozygous for the 1.25-kb *TaqI* RFLP. The Icelandic patients were also homozygous for the 8-kb *SphI* RFLP, but the British patient was heterozygous at this site. These studies suggest that a founder effect is likely to be responsible for APRT deficiency in the Icelandic population. The finding of the same mutation in a patient from Britain suggests that this mutation may have originated in mainland Europe.

Introduction

Adenine phosphoribosyltransferase (APRT; E.C. 2.4.2.7) is a purine-metabolism enzyme that catalyzes the synthesis of AMP from adenine and 5-phosphoribosyl-l-pyrophosphate. In patients with complete APRT deficiency (McKusick 102600), adenine is oxidized, by xanthine oxidase, to the highly insoluble and nephrotoxic derivative, 2,8-dihydroxyadenine (DHA). This produces crystalluria and can lead to kidney-stone formation and eventual renal failure (Simmonds et al. 1989).

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Two phenotypic variants of APRT deficiency have been recognized, but the pattern of clinical expression in the two types is similar. Type II deficiency (complete enzyme deficiency in vivo but partial deficiency in cell extracts) has been found only in Japan. Type II patients are homozygotes or compound heterozygotes for the *APRT*J* missense mutation in exon 5 (Kamatani et al. 1987, 1990*b;* Hidaka et al. 1988; Sahota et al. 1990). Type I APRT deficiency (complete deficiency in vivo or in vitro), on the other hand, has been found in patients from many different countries. Type I patients are homozygotes or compound heterozygotes for a variety of null alleles collectively designated *APRT*Q0* (Fujimori et al. 1985; Hidaka et al. 1987). To date, three different *Q0* mutant alleles have been identified (Hidaka et al. 1987; Sahota et al. 1990).

In the present report we describe the identification of a single missense mutation (GAC→GTC) in exon 3 (asp_{65}→val) in all five patients with type I APRT

deficiency reported from Iceland and in a patient from Britain.

Patients and Methods

Patients

Icelandic cases I–IV have been described elsewhere (Laxdal and Jonasson 1988). Case V is a 42-year-old Icelandic woman whom we have identified recently. Her clinical history is similar to those of most of the previously described adult cases (Simmonds et al. 1989). She had recurrent urinary-tract infections from age 10 years. Her first renal colic occurred at age 23 years, and she has had recurrent attacks since then. Bilateral nephrolithotomy was performed at age 29 years. She passed a right ureteral stone in December 1989, and at that time typical DHA crystals were noticed by microscopy of the urine. The composition of the stone was confirmed by UV and infrared spectroscopy. She was started on allopurinol in March 1990, and, at a dose of 200 mg/d, the urine crystals disappeared.

Cases I (AJ) and II (HH) shared common ancestors six and seven generations ago, and cases I and V (HS) had common ancestors seven generations ago (fig. 1). Cases III (PJ) and IV (HE) are not known to be interrelated or to show any relationship to the other three cases, over at least seven generations. The maternal grandfather of case II is of British descent (fig. 1).

Case VI (SB), now age 14 years, is a patient from Britain and has been described elsewhere (Greenwood et al. 1982). In brief, she presented at age 4 years with renal failure due to DHA lithiasis. She was comatose, anuric, and required peritoneal dialysis. Ultrasonography and retrograde pyelography showed the presence of bilateral renal caliculi. The stones were removed by pyelolithotomy and were identified as DHA. This case is not known to have any relationship to the Icelandic cases.

Methods

Blood collected in sodium heparin–containing vacutainers was sent by air to Indianapolis from Iceland and Britain. B-lymphoblast transformations from the patients and their immediate relatives were initiated within 3 d of blood collection (Neitzel 1986). APRT activity was measured in hemolysates and in lymphoblast extracts (Sahota et al. 1991).

DNA was isolated from blood or from cultured lymphoblasts, and a 2.4-kb fragment spanning both

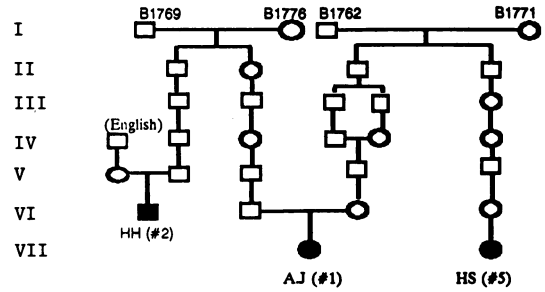


Figure 1 Pedigree of Icelandic cases I, II, and V. The year of birth of the four individuals in generation I is indicated.

the entire coding region and the flanking sequences of the *APRT* gene was amplified by PCR using the Perkin Elmer–Cetus thermal cycler and the GeneAmp kit. The amplification conditions were as follows: 0.5 μg DNA, 0.4 μg each primer (see below), 1.2 mM $MgCl_2$, 0.2 mM each deoxyribonucleotide, 5% dimethylsulfoxide, 1 min denaturation at 95°C, 1 min annealing at 60°C, 4 min extension at 72°C, and 30 cycles. The initial denaturation of the template was for 7 min in a boiling-water bath. Other conditions were as described in the GeneAmp manual. Primers NC720 (sense sequence) and C719 (antisense sequence), located at positions −161 to −140 and 2242 to 2221, respectively, of our wild-type sequence, were used for amplification (Chen et al. 1991). (The A of the ATG start codon is designated base 1.) The primer sequences (5′ to 3′) were ATGGCGCCTAGGAGTCCATGAA (NC720) and TGGGATCCAGCTGGAGATGTTG (C719).

The PCR reaction mixture was electrophoresed on an agarose gel, and the 2.4-kb fragment was purified from the gel by using GeneClean (BIO 101; La Jolla, CA). Approximately 200 ng PCR-amplified DNA was then ligated with 50 ng M13mp18 RF DNA which had been digested with *Hin*cII. After *Escherichia coli* JM109 transformation, single-stranded DNA from three to five positive plaques from each patient was sequenced completely (Chen et al. 1991). The PCR-amplified DNA from each patient was also sequenced directly.

For RFLP analysis, genomic DNA was digested with *Sph*I and PCR-amplified DNA was digested with *Taq*I. The *Sph*I digests were electrophoresed on agarose gels, were transferred to GeneScreen Plus membranes, and were probed with the 2.2-kb *Bam*HI fragment of the *APRT* gene (Stambrook et al. 1984). Prehybridization and hybridization conditions were as

described in the GeneScreen instruction manual. The *Taq*I digests were electrophoresed on agarose gels and stained with ethidium bromide. The polymorphic *Sph*I site lies outside the *APRT* gene and is 2.8 kb upstream (Kamatani et al. 1990*a*) of the polymorphic *Taq*I site, which is located within intron 2 (Chen et al. 1991). Nucleic acid and protein sequences were analyzed using the Microgenie (V5.0) software from Beckman.

Results

APRT activities in hemolysates from the Icelandic patients (I–V), from the British patient (VI), and from immediate relatives are given in table 1. All patients were deficient in enzyme activity; the low level of activity seen in cases II, III, and V is not significantly different from the background. Lymphoblast extracts from the patients showed <1% activity of the control mean (614 nmol/h/mg, SD = 26, n = 6).

An A-to-T transversion at position 1350 (exon 3) was identified in all five Icelandic patients (fig. 2). The sequences of the exon, intron, and flanking regions were otherwise identical to those of the wild-type gene (Chen et al. 1991). The same mutation was found in the patient from Britain. The homozygous nature of the mutation in all six patients was confirmed by direct sequencing of PCR-amplified DNA. This mutation predicts the replacement of asp (GAC) by val (GTC) at amino acid position 65.

Table 1

Patient Details and APRT Activity in Hemolysates

Case or Parent	Name	Sex	Age at Onset (years)	Enzyme Activity[a] (nmol/h/mg Hb)
Icelandic:				
I	AJ	F	3	.0
Mother				15.1
Father				12.6
II	HH	M	43	1.0
Mother				12.4
III	PJ	F	42	0.7
Brother				32.5
Brother				11.7
Brother				14.8
IV	HE	F	1	.0
Mother				10.5
Father				14.2
V	HS	F	42	.5
British:				
VI	SB	F	5	.0
Sister				19.9
Mother				5.4

[a] Mean ± SD range in controls (n = 46): 24.5 ± 4.8.

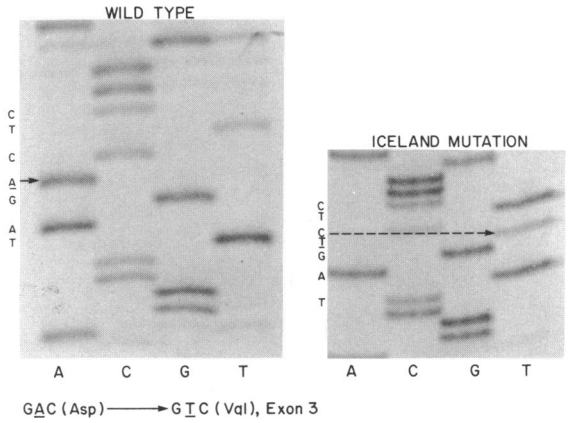


Figure 2 DNA sequence showing single A-to-T substitution at position 1350. This mutation leads to the replacement of asp (GAC) by val (GTC) at amino acid position 65 (exon 3).

All six patients were homozygous for the 1.25-kb *Taq*I RFLP, as observed in digests of PCR-amplified DNA. The digest pattern for three of these patients (Icelandic cases I–III) and for two control subjects is shown in figure 3. The 1.25-kb fragment seen in *Taq*I digests of PCR-amplified DNA is equivalent to the 2.1-kb fragment seen in digests of genomic DNA (Stambrook et al. 1984; Chen et al. 1991). All five Icelandic patients were homozygous for the 8-kb *Sph*I RFLP. The three sons of case III, the mother of case IV, and the British patient (case VI) were heterozygous at this site (fig. 4).

Discussion

An exon 3 missense mutation (GAC→GTC at position 1350, asp_{65}→val) was identified in all five Icelandic patients (I–V) who had APRT deficiency. Patients I and II are related six and seven generations back, and patients I and IV are related seven generations back, (fig. 1), but patients III and IV are not demonstrably related either to any of these three or to each other, over seven generations. It seems most likely, however, that the mutation in all five patients has a common ancestral origin, since most Icelanders are discovered to be interrelated if the genealogy is followed back 12–14 generations. The absence of any sequence variations either within the *APRT* gene or in the immediate

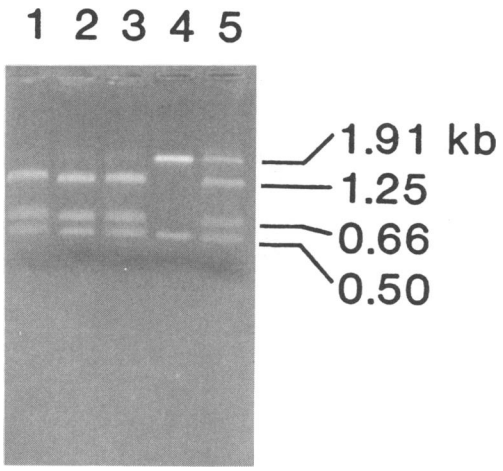


Figure 3 Ethidium bromide–stained agarose gel showing PCR-amplified DNA digested with *Taq*I. Lanes 1–3, Cases I–III, respectively. Lanes 4 and 5, Control subjects. The digestion pattern for cases IV–VI (not shown) was identical to that for cases I–III. The 0.5-kb fragment in all five lanes results from an invariant *Taq*I site in exon 2 (Chen et al. 1991). The 1.91-kb fragment (1.25 + 0.66 kb) in lanes 4 and 5 is due to the loss of the polymorphic *Taq*I site in intron 2 and is equivalent to the 2.8-kb fragment seen in genomic digests (Stambrook et al. 1984).

flanking regions from these patients supports the notion of common ancestry. It also suggests that the mutation may be of relatively recent origin.

The same missense mutation was found in the British patient (VI) but not in four other European patients we have examined (data not shown). Since the maternal grandfather of patient II is of British origin (fig. 1), one of the mutant alleles from this patient could have come from Britain, but it is equally likely that the mutant allele came from the paternal grandfather, who is Icelandic. It is also possible that the mutation in the Icelandic patients and in the British patient are of independent origin. Sufficient family data are not available to differentiate between these possibilities. It is unlikely that the GAC triplet encoding asp_{65} is a hot spot for mutation, since we have not seen this particular mutation in 13 other type I patients from different groups (data not shown). *Sph*I digestion of genomic DNA from all five Icelandic cases gave an 8-kb fragment on a Southern blot. Digestion of DNA from the three sons of case III, from the mother of case IV, and from case VI gave 8- and 12-kb fragments, indicating the presence of both *APRT* alleles (fig. 4). The Icelandic patients are most likely to be homozygotes at this site, but the possibility of a deletion of one of the *APRT* alleles cannot be ruled out.

Previous studies have shown that the *APRT*J* mutation in patients with type II deficiency has a common ancestral origin (Kamatani et al. 1987, 1990*a*; Hidaka et al. 1988). We and others have recently identified a common nonsense mutation in exon 3 in Japanese patients with type I APRT deficiency (Sahota et al. 1990; Mimori et al. 1991). We have also shown that the previously described mutation at the intron 4 splice-donor site in one allele from a Belgian family is a common cause of APRT deficiency in the Caucasian population (Hidaka et al. 1987; Gathof et al., in press). The finding of another common mutation, $APRT_{Iceland}$, suggests that a small number of different mutations is likely to be responsible for the majority of reported cases of APRT deficiency. Unlike some previously reported mutations in the *APRT* gene (Hidaka et al. 1988; Mimori et al. 1991), the Iceland

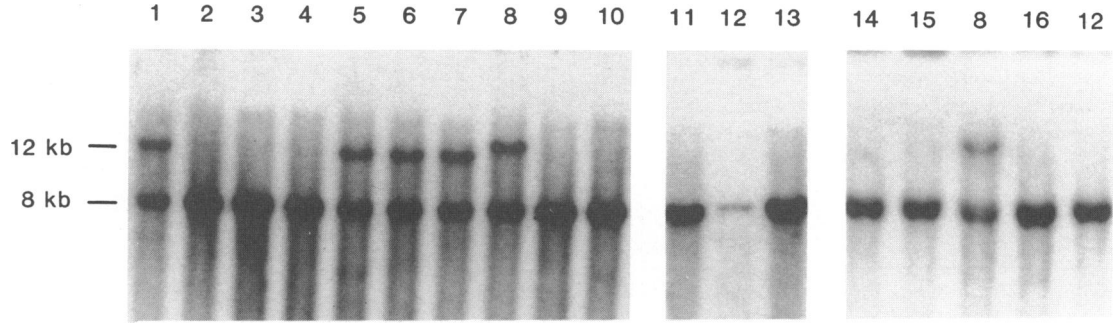


Figure 4 Southern blot showing *Sph*I digestion of genomic DNA. Lane 1, Case VI (SB). Lanes 2 and 3, Mother and father, respectively, of case I. Lane 4, Case III (PJ). Lanes 5–7, Sons of case III. Lanes 8 and 9, Mother and father, respectively, of case IV. Lane 10, Mother of case II. Lane 11, Icelandic control subject. Lane 12, Case V (HS). Lane 13, APRT-deficient human cell line (HTD114). Lane 14, Case I (AJ). Lane 15, Case IV (HE). Lane 16, Case II (HH). Digestion and blotting of DNA was repeated for lanes 8 (mother of case IV) and 12 (case V), and identical results were obtained, as shown in the right-hand side of the figure.

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HU:  Y I A G L D S R G F L
MO:
HA:
DM:  K   V
EC:  I V V   T E A
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Figure 5 Comparison between human APRT amino acid sequence flanking asp_{65}→val mutation and sequences from four other species. Asp_{65} is underlined. Only differences from the human sequence are shown. HU = human; MO = mouse; HA = hamster; DM = *Drosophila melanogaster;* EC = *Escherichia coli.*

mutation does not abolish or create a restriction site for any known restriction endonuclease.

The $APRT_{Iceland}$ mutation occurs in an 11-amino-acid stretch that is fully conserved among human, mouse, and hamster APRTs. This region also shows extensive homology with the enzyme from *Drosophila melanogaster.* The sequence of the *Escherichia coli* enzyme in this region differs from the above sequence at several positions, but the differences are all neutral substitutions (Johnson et al. 1987) (fig. 5). The GAC→GTC mutation, resulting in an asp_{65}→val substitution, is the only change, vis-à-vis the wild-type sequence, that we have identified in the six patients described here. The asp_{65}→val substitution in the enzyme predicts both a change from hydrophilicity to hydrophobicity and a change in secondary structure from β-sheet/random coil/β-turn to β-sheet. The loss of enzyme activity after this substitution suggests that asp_{65} is important for structural or functional integrity of the enzyme.

Acknowledgments

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